

4.3. Discovery

This section describes Service Advertising and Discovery for Matter.

Service Advertising and Discovery is used within Matter in the following contexts:

- [Commissionable Node Discovery](#)
- [Operational Discovery](#)
- [Commissioner Discovery](#)
- [User Directed Commissioning](#)

Service Advertising and Discovery for Matter uses IETF Standard DNS-Based Service Discovery (DNS-SD) [RFC 6763]. Matter requires no modifications to IETF Standard DNS-SD.

Using DNS-SD means that both the unicast IPv6 address and port of the offered service are discovered, freeing Matter from requiring a single preallocated fixed port. This also makes it possible to run multiple instances of Matter software on a single device, because each instance has its own dynamically allocated port, instead of conflicting over attempting to use the same preallocated fixed port.

On Wi-Fi and Ethernet networks today, DNS-SD [RFC 6763] uses Multicast DNS [RFC 6762] for zero-configuration operation.

Since Matter protocols must support IPv6 at a minimum, Matter software discovering other Matter instances SHALL process DNS AAAA (IPv6 address) records, but also MAY process DNS A (IPv4 address) records.

Because of this, where feasible in the underlying service discovery API, Matter software advertising the availability of a service SHOULD indicate that announcements and answers for this service need include only IPv6 address records, not IPv4 address records. On a general-purpose dual-stack host that supports both IPv4 and IPv6, this can be achieved by having Matter-related SRV records reference a Matter-specific target hostname that has only IPv6 address records attached. This allows a general-purpose dual-stack host to offer discoverable IPv4 addresses for legacy client software that still requires IPv4, while offering optimized IPv6-only address discovery for Matter purposes.

Similarly, since Matter does not use IPv4, Matter software discovering other Matter instances SHOULD NOT expect any IPv4 addresses included in responses.

These two items address the *content* of service discovery messages. When using Multicast DNS similar efficiency questions arise related to the *delivery* of those service discovery messages, sent over IPv4, IPv6, or both.

Where supported in the underlying service discovery API, Matter software using Multicast DNS to advertise the availability of a service SHOULD indicate that announcements and answers for this service need only be performed over IPv6.

Similarly, where supported in the underlying service discovery API, Matter application software using Multicast DNS to issue service discovery queries SHOULD indicate that these queries need